QAELUM

ORACLE INTELLIGENCE

CONFIDENTIAL · INVESTOR DISTRIBUTION ONLY

TECHNICAL & INVESTMENT WHITEPAPER · V2.2

An Intelligence Platform for the *Markets Bloomberg* Cannot See.

Ground-truth commodity data from miners on four continents. Frontier-market DEX execution signals. AI model licensing. Proprietary arbitrage as a fourth revenue stream. One Bittensor subnet, four independent income lines.

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Version 2.2

May 2026

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Document *Map.*

⚠ **PRE-MAINNET DISCLOSURE**

QAELUM Oracle Intelligence is currently a documented architecture with unaudited smart contracts and no live mainnet operations. This whitepaper describes a system in development, not in production. Forward-looking statements represent founder intent rather than guarantees. All seed capital is at risk of total loss.

SECTION 01

Executive *Summary.*

QAELUM Oracle Intelligence is being built as an institutional-grade intelligence platform for frontier and emerging markets, with proprietary arbitrage as one of four revenue streams rather than the central thesis.

The Core Thesis

Bloomberg, Refinitiv, and S&P Global Commodity Insights are excellent at what they do for developed markets. None of them serve the markets QAELUM is designed for. Cocoa farmgate dynamics in Ondo State, Naira-USDT corridor spreads in Lagos, USDD-USDT depeg patterns on Tron, Argentine peso devaluation rates against DAI: these are markets institutional data providers either cannot reach or have structurally chosen to ignore. The intelligence gap is real, persistent, and increasingly monetisable as global capital flows shift toward emerging-market allocation.

QAELUM closes this gap through a Bittensor subnet that coordinates ground-truth miners across four continents, fine-tunes a 7-billion-parameter financial language model on data that cannot be purchased, and packages the resulting intelligence into subscription APIs and AI licensing products. The arbitrage execution layer monetises the same data internally; it is a revenue stream, not the platform's reason for existence.

Why The Intelligence Platform Anchors Valuation

Pure-data and intelligence companies in the crypto and frontier-finance sector command premium valuations: Kaiko at $200M+, Nansen at $750M peak, Glassnode at $100M+, Coin Metrics at $50-100M, Amberdata at $200M+. None of these companies run proprietary arbitrage. None have ground-truth commodity data from frontier markets. None have a Bittensor-native distribution model. All are valued at orders of magnitude above QAELUM's $2.5M seed valuation.

Under a stress-test scenario in which QAELUM's arbitrage business generates zero profit, the intelligence platform alone (QSIS, QFSS, QCIS, QAELUM-FIN licensing) is projected to reach $3M ARR by Year 2. At the conservative 10x ARR multiple typical of data infrastructure businesses, QAELUM has a defensible valuation floor of $30M even with complete arbitrage failure. The arbitrage business compounds returns when it works. The data platform anchors valuation regardless.

Why Institutional Quant Firms Cannot Replicate This

Citadel, Two Sigma, DE Shaw, and Renaissance Technologies have substantially more compute, capital, and talent than QAELUM. They will continue to dominate developed-market financial intelligence. They will not enter QAELUM's target markets because their compliance frameworks, regulatory exposure, and capital deployment models are incompatible with frontier-market operations. They cannot have miners in Lagos. They cannot operate on Tron. They cannot custody Naira at scale. QAELUM-FIN does not compete with institutional quant firms in their core markets; it serves markets they are structurally barred from. That is the entire moat.

STAGE	TARGET RAISE	PRE-MONEY	REVENUE LINES
Seed	$750K	$2.5M	4

Pre-revenue, pre-mainnet	SAFE · cap $850K	Defensible floor	Independent streams

SECTION 02

Two Real Problems. *Both Underserved.*

Problem 1: The Frontier Intelligence Gap

Cocoa traded for $4,800 per metric ton on ICE in March 2025. The actual farmgate price paid to Ghanaian and Nigerian farmers ranged between 65 and 75 percent of that figure, with material variation across regions and weeks. A trader who knew the actual farmgate dynamics had an information edge that the futures market would only price in 18-72 hours later. That information edge is currently captured by a small group of physical commodity traders (Cargill, Olam, Wilmar) and is unavailable to financial market participants who would pay substantial subscriptions to access it.

This is one example. The same gap exists for cotton in West Africa, palm oil in Indonesia, lithium in DRC, coffee in Vietnam, and dozens of other commodity-currency pairings. The institutional buyers exist (hedge funds with EM mandates, family offices with agricultural exposure, commodity-focused funds). The institutional sellers do not exist — at least not at sufficient scale or with sufficient verification rigor to satisfy institutional procurement.

Problem 2: Frontier-Market DEX Inefficiencies

Decentralised exchanges in frontier markets exhibit price divergences that would be arbitraged away in seconds on Ethereum or Solana. USDD on SunCurve consistently depegs from USDT. PSM redemption mechanisms allow 1:1 conversion at zero fee. Stablecoin pairs across Tron, Arbitrum, Base, and Optimism show 0.04-0.4 percent spreads multiple times per day. These opportunities persist because institutional MEV operators (Wintermute, GSR, Flow Traders) cannot deploy at scale into pools that are too shallow for their capital sizes. The structural alpha floor exists; capturing it requires operational presence the institutional players lack.

> **★ WHY THESE TWO PROBLEMS COMBINE**
>
> The same Bittensor subnet infrastructure that coordinates QCIS commodity miners can simultaneously coordinate QAE arbitrage path discovery. The same QAELUM-FIN model that ingests QCIS data can output QSIS signals. The same smart contract that emits proof-of-alpha can verify commodity data submissions. Shared infrastructure across two large addressable markets is what makes the platform economics work.

SECTION 03

Four Independent *Revenue Lines.*

Each line can support QAELUM independently in adverse scenarios. Together they compound. The intelligence platform leads; arbitrage follows.

Line 1 — QCIS: Commodity Intelligence API

Ground-truth frontier-market commodity data sold to institutional buyers. Hedge funds with EM mandates, commodity trading houses, agricultural finance firms, treasury teams, and academic researchers subscribe to verified data feeds that no institutional provider offers.

Pricing tiers: Basic ($499/mo), Institutional ($2,499/mo), Enterprise ($9,999/mo), Premium with custom miner sourcing ($19,999/mo).

Target customers Year 2: 25-50 institutional subscribers. **Projected ARR Year 2:** $750K-$2M.

Line 2 — QAELUM-FIN: AI Model Licensing

A 7-billion-parameter language model fine-tuned on a proprietary dataset that institutional firms cannot purchase. Licensed to other Bittensor subnets that consume financial intelligence, to academic researchers, and to institutional buyers integrating into proprietary trading systems.

Pricing: Combination of subscription access ($25K-$100K annual enterprise contracts) and per-inference pricing for high-volume consumers.

Projected ARR Year 2-3: $400K-$2M. The licensing line has the highest gross margins (approaching pure-software economics) once the model is trained.

Line 3 — QSIS / QFSS: Signal APIs

DEX execution signals (QSIS) and commodity futures predictions (QFSS) packaged as subscription APIs. Different products from QCIS — these are derived predictive signals rather than raw data — sold to a different customer base (quant desks, prop shops, retail platforms).

Pricing tiers: Retail ($499/mo), Institutional ($2,499/mo), Enterprise ($9,999/mo).

Projected ARR Year 2: $800K-$1.6M.

Line 4 — QAE: Proprietary Arbitrage

Atomic flash loan arbitrage cycles across Tron, Arbitrum, Base, and Optimism. Net profit distributed 40/30/20/10 (investors/miners/burn/treasury) via smart contract enforcement.

Phase 1 projection: 50-150 cycles/day, $30-$120 average net profit per cycle, $500K-$5M annual gross revenue.

Strategic role: Funds operations during pre-revenue scaling. Provides direct yield to seed investors. Generates additional training data for QAELUM-FIN. Not the valuation driver.

Line 5 (Bonus) — Bittensor TAO Emissions

As a registered Bittensor subnet, QAELUM earns TAO from network emissions proportional to validator-assigned trust scores. This is denominated in TAO, exposing this stream partially to TAO price movement. **Crucially, this is the only stream affected by TAO price; the other four are USD/USDT-denominated and operate regardless of crypto market conditions.**

> ★ **STRESS TEST: ARBITRAGE FAILURE SCENARIO**
>
> If Line 4 generates zero profit for two consecutive years, QAELUM still has Lines 1, 2, 3, and 5. Conservative projections show $3M ARR by Year 2 from those four lines alone. At 10x ARR multiple (standard for data infrastructure), the company is worth $30M even with complete arbitrage failure. The diversified revenue architecture is not theoretical risk mitigation — it is the structural reason the platform is defensible.

SECTION 04

QAELUM-FIN: *The AI Layer*.

A fine-tuned 7-billion-parameter language model whose competitive advantage is not the architecture (publicly available) but the training corpus (impossible to purchase).

Model Architecture

Base model: Mistral 7B (Apache 2.0 license). Fine-tuning method: LoRA (Low-Rank Adaptation) for parameter-efficient specialisation. Inference deployment: 4-bit GGUF quantisation reduces memory footprint to ~6GB VRAM, allowing deployment on consumer-grade hardware (RTX 3060 and equivalent). Distributed inference is performed by Bittensor validators rather than centralised servers.

The Training Corpus (Where the Moat Lives)

The model architecture is replicable in 4 hours by anyone with cloud access and basic ML engineering skills. The training corpus is structurally not replicable. It contains:

- **Founder's documented arbitrage attempt logs (2020-2026):** success and failure outcomes across DEX execution attempts, with slippage data, MEV impact analysis, and gas economics.
- **Real DEX execution outcomes:** historical and live data across 66 monitored pairs covering Tron, Arbitrum, Base, Optimism, and Polygon.
- **QCIS commodity submissions:** verified ground-truth pricing data from QCIS miner network across cocoa, cotton, coffee, palm oil, lithium, cobalt, and crude oil markets.
- **Fiat corridor dynamics:** spread observations from Nigerian Naira, Argentine peso, Turkish lira, and Vietnamese dong corridor pairs against major stablecoins.
- **Bittensor on-chain proof history:** verified subnet activity providing cross-subnet intelligence not available outside the network.

This corpus takes 5+ years of operational engagement to assemble. A competitor with $50M in funding could spend 18-24 months building a comparable miner network and dataset; in that time, QAELUM compounds its data advantage further. The lead is sustainable.

Why Citadel Cannot Build This

It is essential that investors understand why institutional quant firms with vastly more resources will not enter QAELUM's space. The barriers are structural, not technical:

- **Regulatory compliance:** Citadel cannot operate on Tron due to Justin Sun's SEC exposure. They cannot custody Naira meaningfully due to Nigerian capital controls. They cannot deploy capital to Argentine peso pairs due to AML/KYC requirements that frontier markets do not satisfy.
- **Capital scale incompatibility:** a $50M deployment into a $30M Tron pool moves the price 5-7% against itself. Citadel cannot meaningfully participate at QAELUM's trade sizes ($50K-$500K).
- **Reputational risk:** association with frontier-market operations carries reputational costs that institutional LPs do not accept. Citadel will not put cocoa farmgate intelligence in their portfolio because explaining it to Yale's endowment is not worth the marginal revenue.

- **Operational model:** Citadel hires 50 quants in Chicago. They do not build distributed miner networks across four continents. The operational model is incompatible with their cost structure and management overhead.

The conclusion: QAELUM-FIN does not need to outperform Citadel's models in absolute terms. It needs to be the only credible model in markets Citadel structurally cannot enter. That is a substantially easier competitive position than direct combat with institutional quant firms.

Comparable Company Valuations

To anchor QAELUM-FIN's potential, comparable companies in the crypto and frontier-finance data sector:

COMPANY	DESCRIPTION	VALUATION
Kaiko	Crypto market data and API	$200M+
Nansen	Wallet intelligence and on-chain analytics	$750M (peak)
Glassnode	On-chain analytics	$100M+
Coin Metrics	Crypto data and indices	$50-100M
Amberdata	Institutional crypto data	$200M+
Messari	Crypto research and intelligence	$40-60M
QAELUM (seed)	Frontier-market intelligence + ground-truth data + AI licensing	**$2.5M (pre-money)**

None of the comparable companies have ground-truth commodity data from frontier markets. None have a Bittensor-native distribution model. None can offer the unique data combination QAELUM is structured to deliver. The seed valuation is conservative even relative to a defensible Year 2 floor of $30M (10x ARR).

Use Cases Driving Licensing Revenue

Three primary use cases for QAELUM-FIN licensing:

1. **Cross-subnet intelligence licensing on Bittensor.** Other subnets (forex prediction, sports prediction, deep research) pay QAELUM for inference access to enrich their own models with frontier-market context.
2. **Hedge fund proprietary integration.** EM-focused hedge funds license QAELUM-FIN as a component of their internal model stack for currency corridor and commodity exposure analysis.
3. **Academic and policy research.** Universities and policy institutes (Brookings, Center for Global Development, IFPRI) pay reduced rates for research access. This builds citation pedigree that compounds into institutional credibility.

> ★ **THE LONG-TERM STRATEGIC POSITION**
>
> QAELUM's terminal goal is to become the standard intelligence layer for capital flowing into frontier markets — what Bloomberg became for developed markets in the 1980s-1990s. The frontier-market capital allocation thesis (BRICS expansion, EM rotation, dollar de-anchoring) is multi-decade in scope. Being early to build the data infrastructure for that capital is structurally a 10-20 year opportunity, not a crypto-cycle opportunity.

SECTION 05

QCIS: The *Defensible Moat.*

A miner in Ondo State submits "cocoa farmgate price today: 8,500 NGN/kg". How does a validator in Singapore verify this without sending someone to Ondo? QCIS answers this question through a four-layer scoring system that makes accurate submission the only profitable strategy.

The Four-Layer Verification Stack

Layer 1 · Anchor Reconciliation (35%)

Each submission is checked against a ground-truth anchor published with delay. The miner submits at time t. The validator stores the submission with BLAKE3 hash. When the anchor publishes at t+24h, the validator computes accuracy_score = 1 - min(1, |miner_value - anchor_value| / tolerance_band). Tolerance bands are commodity-specific (cocoa: 3%, cotton: 4%, coffee: 2.5%, lithium: 5%, crude: 1.5%).

Layer 2 · Cross-Miner Consensus (25%)

When multiple miners submit values for the same commodity within a coordinated window, their submissions are clustered. The median becomes consensus. Miners within 1σ of cluster median: full score. 1-2σ: half score. Beyond 2σ: zero or negative. Critical retrospective correction: a miner whose submission was an outlier at consensus time but who is later validated by Layer 1 anchor publication receives substantial retroactive rewards.

Layer 3 · Temporal Consistency (20%)

Commodity prices change in plausible distributions. The plausibility score = exp(-((change_rate - historical_mean) / 2σ)²) where the historical baseline is commodity-specific. Miners submitting realistic day-over-day variations score high. Wild swings (8,500 → 12,000 → 4,000) score low regardless of individual accuracy.

Layer 4 · Behavioural Signature (20%)

Sybil resistance and bot detection. Real human miners exhibit timing patterns (local business hours), frequency variation (miss days for harvests and holidays), geographic clustering around real markets, natural submission entropy, and verified hardware attestation. Miners failing behavioural fingerprinting score lower regardless of accuracy.

```
COMPOSITE REPUTATION SCORE

R(t) = 0.35 · anchor_accuracy
    + 0.25 · consensus_alignment
    + 0.20 · temporal_plausibility
    + 0.20 · behavioural_signature

R_total(t) = 0.15 · R(t) + 0.85 · R_total(t-1)
(exponential moving average — recent matters more, history preserved)
```

Reputation Tiers and Slashing

Miners with R_total above 0.85 receive verified status; their submissions carry institutional weight. Miners between 0.60-0.85 receive proportional rewards. Below 0.40 receive zero rewards. Below 0.20 for 30 days triggers automatic deregistration. Miners detected submitting provably false data (anchor reconciliation showing >5σ error) face immediate 50% reputation slash, 7-day payment suspension, exclusion from QAELUM-FIN training corpus, and permanent ban with staked TAO slashing for repeat offences.

Defeating Specific Gaming Strategies

ATTACK STRATEGY	WHY IT FAILS
Submit random values	Layer 1 anchor reconciliation slashes upon anchor publication
Copy other miners	Layer 2 detects lockstep patterns; Layer 4 fingerprinting exposes sybil clusters
Bot that mimics anchor + noise	Anchor publishes after submission window; attacker has nothing to copy at submission time
Match public Bloomberg data	Bloomberg lags by hours and lacks frontier coverage; matching provides zero predictive value
Multiple identities (sybil)	Hardware attestation + Layer 4 behavioural analysis links identities
Submit only easy days	Layer 3 penalises gap patterns; low frequency reduces reputation accumulation
Bribe an insider for early anchor	Suspiciously high accuracy from small miner subsets triggers anomaly detection
Build competing miner network using QAELUM API output	Competing network must outscore QAELUM miners in same scoring system, requiring more accurate ground truth, requiring actual regional presence — attack reduces to building QAELUM from scratch

The Compounding Network Effect

A miner with R_total of 0.91 sustained over 90 days has demonstrated genuine ground-truth presence verified through four independent layers. That miner's reputation is itself a marketable asset; institutional buyers paying QCIS for "verified miner identity" want this specific miner's data, not a newcomer's. A competitor launching a copy QCIS faces a problem they cannot solve quickly: zero miners with track records, zero anchor history, zero consensus alignment data, no reputation database. They cannot fast-forward 90 days of operation. The smart contracts can be forked in a weekend. The miner network cannot.

From Verified Data to Institutional Revenue

QCIS data monetises through four channels: direct API subscription (hedge funds, commodity trading houses, treasury teams), QFSS predictive signal layer (derivative product at higher margin), QAELUM-FIN training input (model licensing revenue partially attributable to QCIS quality), and Reputation Oracle (cross-subnet revenue where other Bittensor subnets pay to verify miner trustworthiness). The same dataset generates revenue four times.

> ★ **WHY THE WILLINGNESS TO PAY EXISTS**
>
> Institutional commodity traders currently pay $5,000-$40,000 per month for satellite-derived crop intelligence (Planet Labs, Maxar, Bayer Climate FieldView). They pay because the data has real economic value. QCIS delivers a different category of data — ground-truth pricing rather than satellite imagery — at

> comparable or lower price points. The institutional budget for commodity intelligence already exists. QCIS captures a portion of that existing budget with complementary, not competing, data products.

SECTION 06

Named Verification *Sources.*

Every QCIS anchor is publicly nameable, externally verifiable, and independently auditable. This section discloses the full source list so investors and partners can validate the system before committing capital.

Primary Government and Trade Body Anchors

Cocoa

Ghana Cocoa Board (COCOBOD) farmgate prices; International Cocoa Organization (ICCO) daily indicator; Côte d'Ivoire Conseil du Café-Cacao (CCC) prices.

Cotton

Cotton Corporation of India (CCI) daily prices; National Cotton Association of Nigeria (NACOTAN) regional prices; Central Bank of Nigeria cotton statistics; Cotlook A Index (international benchmark).

Coffee

International Coffee Organization (ICO) Composite Indicator Price; Brazilian National Coffee Council (CNC) prices; Coffee Quality Institute regional benchmarks.

Palm Oil

Malaysian Palm Oil Board (MPOB) daily settlements; Indonesian Palm Oil Association (GAPKI) prices; Bursa Malaysia Derivatives futures settlements.

Crude Oil & Energy

OPEC Reference Basket pricing; Nigerian Department of Petroleum Resources (DPR) export prices; Argus Media benchmarks; S&P Global Platts.

Critical Minerals (Lithium, Cobalt, Rare Earths)

Benchmark Mineral Intelligence; Fastmarkets; London Metal Exchange (LME) settlements; Shanghai Metals Market (SMM).

Precious Metals

London Bullion Market Association (LBMA) fixing; COMEX settlements.

Secondary Cross-Validation Anchors

Futures Markets

Intercontinental Exchange (ICE): Cocoa, Cotton 2, Coffee C, Sugar 11, Robusta. Chicago Mercantile Exchange (CME): commodity contracts and indices. Bursa Malaysia Derivatives. Shanghai Futures Exchange (SHFE).

Trade and Customs Data

UN Comtrade database (international trade statistics); ITC Trade Map (export volumes); World Bank Commodity Markets Outlook (Pink Sheet).

Satellite and Agricultural Data

NASA MODIS (vegetation indices, free public access); ESA Sentinel-2 (high-resolution Earth observation); USDA Foreign Agricultural Service crop estimates.

Logistics and Shipping Cross-References

Drewry Container Index; Baltic Dry Index; port congestion data (Lloyd's List Intelligence).

Weather and Climate Anchors

NOAA Climate Prediction Center; European Centre for Medium-Range Weather Forecasts (ECMWF); NASA Earth Observatory.

Currency and Fiat Corridor Data

Federal Reserve Economic Data (FRED); Central Bank of Nigeria official rates; Banco Central de la República Argentina; Türkiye Cumhuriyet Merkez Bankası; State Bank of Vietnam.

★ **AUDITABILITY**

Every anchor named above is publicly published, externally auditable, and citable in QAELUM's training data documentation. The combination of multiple independent sources is what makes the system resistant to single-source manipulation, as detailed in the following section.

SECTION 07

When Anchors *Lie.*

Government and institutional data sources do sometimes publish manipulated values — for policy reasons, economic management purposes, or simple error. QCIS is specifically designed to handle this case. The lying anchor problem becomes a product feature, not a vulnerability.

How Lying Anchors Get Handled

Consider the realistic scenario: COCOBOD publishes a cocoa farmgate price of 7,800 NGN/kg for policy reasons, while the actual market price observed by miners across multiple regions is 8,500 NGN/kg. A naive single-anchor scoring system would punish the honest miners. QCIS handles this through three layered defences:

Defence 1 — Multi-Anchor Triangulation

No single anchor controls scoring. For cocoa specifically: COCOBOD, ICCO, ICE Cocoa Futures, and miner consensus are all evaluated together. Three of four sources must broadly agree for an anchor to score normally. When COCOBOD diverges from ICCO + ICE + consensus, the system identifies COCOBOD as the outlier rather than punishing miners who match the broader truth.

Defence 2 — Anomaly Detection Override

When the published anchor diverges from miner consensus by more than 4 standard deviations, the system flags an anchor anomaly event. Four things happen automatically:

1. The anchor's weight in scoring drops from 35% to 10% for that period
2. The cross-miner consensus weight rises from 25% to 50%
3. A retrospective audit window opens; if the anchor publishes a correction within 30 days, miner scores are recomputed
4. The anchor itself accumulates reputation damage in QCIS's internal scoring of anchor sources

Defence 3 — Layer Composition

Even without anomaly detection, Layer 1 (anchor reconciliation) is only 35% of total score. The other 65% (consensus, temporal, behavioural) is anchor-independent. A miner who truthfully reports 8,500 NGN/kg while COCOBOD lies at 7,800 receives a reduced Layer 1 score (~10.5% partial) but maintains full Layer 2/3/4 scores (~60% total). Composite reputation stays at ~70%, still in verified tier.

> **★ THE LIE BECOMES ALPHA**
>
> The strategic insight: when Bloomberg or COCOBOD publishes a manipulated value, the divergence between published and ground-truth pricing is itself a tradeable signal. Hedge funds pay premium prices for "official source says X, ground truth says Y, position accordingly" intelligence. QAELUM does not need to choose between trusting the anchor and trusting the miners. It reports both, flags the divergence, and lets institutional subscribers trade against the manipulation. The vulnerability becomes the product.

The Anchor Reputation System

Each anchor source in QCIS has its own reputation score, recalculated continuously. Anchors that consistently align with the four-way triangulation maintain high weight. Anchors that systematically diverge (whether due to manipulation, methodology change, or institutional policy shift) lose weight. New anchors can be added to the triangulation through governance proposals as new sources become available. The system is adaptive, not static.

Why This Matters For Investor Confidence

A common concern from sophisticated investors: "What stops manipulated official data from corrupting the entire system?" The answer is that QCIS is structurally designed to detect manipulation as a feature rather than fall victim to it as a bug. Every layer of the scoring system reinforces this: multi-source triangulation prevents single-source dominance, anomaly detection auto-reweights during divergence events, anchor reputation tracking creates feedback loops that punish unreliable sources, and the lie-as-alpha business model converts manipulation into revenue. This is not a system designed to trust institutions blindly. It is a system designed to verify institutions through cross-reference.

SECTION 08

Signal *APIs.*

QSIS and QFSS package the intelligence layer into subscription products for quantitative traders and institutional desks.

QSIS — QAELUM Signal & Intelligence System

Real-time DEX execution signals derived from the multi-DEX liquidity scanner (MDLS) and SQBM path optimisation engine. Subscribers receive feeds containing: pair-specific spread signals across 66 monitored DEX pairs, slippage and execution-cost predictions, MEV-vulnerability flags, gas-cost forecasts, and confidence scores per signal. Designed for quantitative traders running their own execution infrastructure who need pre-trade intelligence on DEX conditions.

Target customers: independent quant desks, prop trading shops, market-making firms with DeFi exposure, retail trader platforms aggregating signals.

Pricing tiers: Retail ($499/mo, 1 chain, basic feeds), Institutional ($2,499/mo, all chains, full feeds), Enterprise ($9,999/mo, custom integrations, dedicated support).

QFSS — QAELUM Futures Signal System

Commodity futures predictive signals derived from QCIS ground-truth data and macro signal layers. Subscribers receive feeds on: commodity-specific futures price direction predictions with confidence intervals, divergence flags between official anchors and ground-truth pricing (the "lie alpha" signal), supply/demand imbalance indicators from logistics and shipping cross-references, weather-derived crop yield adjustments.

Target customers: commodity-focused hedge funds, agricultural trading houses, family offices with commodity exposure, ESG funds tracking commodity supply chains.

Pricing tiers: Standard ($2,499/mo), Enterprise ($9,999/mo with custom commodity coverage), Premium ($19,999/mo with direct miner sourcing).

Distribution Strategy

Both APIs distributed via standard REST and WebSocket interfaces with rate limits per tier. Authentication via API keys with hardware token 2FA for institutional tier. Latency targets: 250ms median for retail tier, 50ms median for institutional tier (dedicated regional endpoints), 20ms median for enterprise tier (private connectivity).

Subscribers can also receive feeds via Bittensor validator nodes for trust-minimised delivery, eliminating QAELUM as a single point of failure for institutional clients with high-availability requirements.

Unit Economics

API products have favourable gross margin profile (approaching 90% at scale). Marginal cost per additional subscriber is dominated by support and infrastructure scaling, not compute. As subscriber count grows, contribution margin per dollar of revenue increases. Year 2 conservative projection: 100-250 subscribers across both APIs, blended ARR of $1.5-3M, gross profit contribution of $1.3-2.7M.

SECTION 09

QAE: The Fourth *Revenue Stream.*

Proprietary arbitrage funds the operation and provides direct yield to seed investors. It is a real business with real economics. It is not the platform's reason for existence.

How It Works

Atomic flash loan arbitrage cycles. The off-chain bot detects price divergence across DEX pairs. The SQBM optimiser selects non-conflicting paths. The smart contract requests a flash loan from AAVE V3 (or Balancer V3, JustLend on Tron). All swaps execute in one transaction. If profit threshold is not met, the entire transaction reverts (capital at risk per cycle: only the gas fee). On success, profit is distributed 40/30/20/10 (investors/miners/burn/treasury) with cryptographically verifiable on-chain proof.

Competitive Position

QAE does not compete with Wintermute, GSR, Flow Traders, or Citadel. Their typical trade sizes ($10M+) are too large for the shallow pools where QAELUM operates. They generate profit at scales QAELUM cannot reach; QAELUM generates profit at scales they cannot reach. The competition is at the edge of overlap (trade sizes in the $500K-$2M range), and even there the regulatory frictions affecting institutional operators on Tron and frontier-market DEXes provide structural protection.

Realistic Performance Targets

Conservative Phase 1 projection: 50-150 successful cycles per day across active chains, $30-$120 average net profit per cycle. Annual gross revenue range: $500K-$5M Year 1. Cycles that fail the profit threshold cost only gas (~$0.04 on Tron, ~$0.50 on Arbitrum), so failed-cycle drag on unit economics is minimal.

Phase 2 multi-chain expansion targets 200-600 daily cycles. Phase 3 with full chain coverage and mature SQBM tuning targets 800-2,000 daily cycles. Revenue scales sub-linearly with cycle count due to liquidity constraints, not linearly with marketing claims.

Why Arbitrage Is Not The Headline

Arbitrage is competitive, volatile, and dependent on factors outside QAELUM's control (gas costs, MEV competition, liquidity depth, sequencer behaviour). It produces variable revenue. It cannot anchor an institutional investment thesis because the unit economics are too dependent on conditions QAELUM cannot guarantee.

The intelligence platform produces recurring, contracted, USD-denominated revenue with 80-90% gross margins and structural moats around training data and miner networks. That is what anchors valuation. Arbitrage is the cash-generation layer that funds the platform build; the platform is what investors are buying.

SECTION 10

Smart Contract *Architecture.*

Engineered Safety Properties

- **Atomic execution:** all swaps occur within one transaction. Partial failure reverts the entire cycle.
- **Daily loss cap:** if 2% of treasury is lost in 24 hours, contract auto-pauses.
- **3-of-5 multisig owner:** contract ownership requires three of five signatures for administrative action.
- **Emergency withdraw requires prior pause:** two-step protection prevents single-signature drain.
- **Custom error types:** all reverts use named errors for gas efficiency and audit clarity.
- **Permit2 approvals:** token approvals scoped to specific amounts and execution windows.
- **Constructor non-zero checks:** all addresses validated at deployment.
- **BLAKE3 hash commitment before distribution:** proof emitted before fund movement, providing audit trail.
- **ML-DSA signature anchoring:** post-quantum signature commitments stored on-chain for long-term verifiability.

Audit Strategy

1. **Pre-audit Immunefi bug bounty ($25K pool):** catches obvious issues cheaply before formal audit.
2. **Spearbit audit ($30-50K):** primary security review by established DeFi auditor. 2-4 week engagement.
3. **Code4rena contest ($25-50K prize pool):** crowdsourced audit by competitive security researchers.
4. **Post-audit remediation (4-6 weeks):** findings remediated, re-audited before mainnet deployment.

Total audit budget allocated: $90-100K of seed proceeds. Audit timeline directly determines mainnet launch date.

Specific Attack Vectors Mitigated

ATTACK VECTOR	MITIGATION
Sandwich attacks	Flashbots Protect on EVM chains; direct SR submission on Tron
Reentrancy	Checks-effects-interactions pattern; ReentrancyGuard on external functions
Oracle manipulation	Multi-source median pricing; no single-pool dependency
Approval drain	Permit2 with scoped time-limited approvals
Token quirks (USDT)	Approval-to-zero before approval-to-amount; fee-on-transfer detection
Sequencer censorship	Multi-chain deployment redundancy
Gas wars / race conditions	minimumProfit threshold reverts on insufficient profit

ATTACK VECTOR	MITIGATION
Compiler bugs	Pinned Solidity version 0.8.24; audit verifies no known CVE exposure

SECTION 11

Multi-Chain *Phased Strategy.*

Each chain has distinct economic and operational characteristics. QAELUM's chain selection prioritises win rate and operational defensibility over raw market hype.

CHAIN	PHASE	BLOCK TIME	GAS/CYCLE	WIN RATE TARGET
Arbitrum (mainnet launch)	Phase 1	250ms	$0.30-1.50	35-50%
Tron (highest win rate)	Phase 2	3,000ms	$0.004	60-80%
Base	Phase 2	2,000ms	$0.10-0.50	40-55%
Optimism	Phase 3	2,000ms	$0.20-1.00	30-45%
Polygon	Phase 3 (evaluate)	2,000ms	Variable	25-35%

Why Arbitrum First

Most mature DeFi tooling, established AAVE V3 with deep flash loan liquidity, Flashbots Protect available, predictable gas pricing, single sequencer reduces MEV variance. Best chain to learn on without expensive mistakes.

Why Tron Second

Highest win rate potential due to architectural privacy (no public mempool prevents external front-running). $0.004 gas per cycle makes even small spreads profitable. PSM stable arb is structurally available. JustLend offers flash loans. Direct relationships with specific Super Representatives enable private submission bypassing the public RPC.

Why Not More Chains Faster

Each chain adds operational complexity. Each requires its own monitoring infrastructure, RPC providers, MEV defence strategy, and audit coverage. Teams that succeed at multi-chain MEV (Wintermute, Flow Traders) deploy 50+ engineers per chain. With contractor engineering capacity and a small team, focused excellence on 2-3 chains beats spread-thin coverage of 8-10.

> ★ LONG-TERM CHAIN STRATEGY
>
> Phase 3+ consideration: native Bittensor EVM deployment. Bittensor's Substrate-based chain added EVM compatibility in 2025. Operating natively on Bittensor's EVM layer eliminates bridge risk for TAO-

denominated revenue, integrates directly with subnet validators, and lowers friction for cross-subnet payments. This is the natural long-term home for QAELUM contracts once Bittensor EVM matures.

SECTION 12

Tokenomics and *Roadmap.*

QAE Token Structure

Total supply: 500 million QAE, fixed, no inflation, no minting authority. Two independent burn mechanisms: 5% of all API revenue and 2% of every arbitrage cycle profit used for open-market buyback with purchased tokens permanently destroyed. Structurally deflationary from launch.

ALLOCATION	%	VESTING
Miner Rewards (TAO emissions reinvestment)	40%	Linear over operational period
Investor Pool (Pro-rata arbitrage distribution)	20%	Continuous from SAFE signing
Team and Founding Contributors	12%	Long cliff with multi-year vest
Protocol Treasury	10%	Governance-controlled
Ecosystem and Partnerships	10%	Milestone-based release
Public Sale / DEX Liquidity	8%	TGE (Phase 3)

Roadmap

Phase 0 — Foundation (Pre-Seed Close)

Close $750K seed, UAE Freezone registration, SAFE execution, Immunefi bug bounty posted, 5 founding miners identified.

Phase 1 — Ship It (Months 1-6)

Testnet deployment, audit completion, QCIS miner network bootstrap to 20+, QSIS/QFSS API beta, mainnet on Arbitrum, first arbitrage cycle, first investor distribution.

Phase 2 — Scale What Works (Months 7-14)

Bittensor mainnet registration, multi-chain arbitrage (Tron, Base), QCIS expansion to 50+ miners, institutional customer onboarding, QAELUM-FIN v1 release.

Phase 3 — Category Definition (Months 15-24)

Series A at ARR-based valuation, VASP licensing (UAE/Mauritius), full-time team expansion, Tier 1 CEX listing for QAE, strategic decision point.

SECTION 13

Risk *Disclosures.*

⚠ **INVESTMENT RISK WARNING**

QAELUM Oracle Intelligence is a pre-revenue, pre-deployment project. Capital invested at seed stage is at risk of total loss. The risks listed below are material and could individually or collectively result in the failure of the venture.

Technical Risks

Smart contract exploit (despite audit), SQBM production-scale failure, Bittensor protocol risk, cross-chain bridge risk, ML-DSA implementation immaturity, oracle manipulation vectors, off-chain bot reliability.

Market and Competitive Risks

Established MEV competition entry, subnet competition on Bittensor (256-slot expansion), crypto market cyclical risk, stablecoin instability (USDD, USDT depeg events), shift in institutional commodity data willingness-to-pay.

Operational Risks

Single-founder execution risk (most material). Talent acquisition risk for contract Solidity, Python, Bittensor specialists. Miner network bootstrap risk if early West Africa contacts do not activate. Audit timeline unpredictability. Founder health, availability, or personal crisis.

Regulatory Risks

VASP licensing complications in UAE/Mauritius. Securities law evolution affecting QAE token characterisation. Fiat corridor restrictions in operating jurisdictions. OFAC compliance complexity across multiple markets. African data sovereignty regulations.

Token-Specific Risks

QAE illiquidity pre-TGE. Token price volatility post-TGE potentially below seed valuation. Vesting cliff restrictions on investor access. TGE delay risk if Phase 3 milestones not met.

Reputational and Trust Risks

QCIS miner network quality control (single bad-actor miner could damage data feed credibility). Public arbitrage failures during early operation could undermine investor confidence. Adverse media coverage of frontier-market crypto operations.

SECTION 14

Founder *Background.*

Nahum — Founder and Chief Architect

I got into crypto in 2020, originally promoting Pi Network on Twitter. That was the entry point, not the destination. From there I started running my own arbitrage experiments across exchanges and chains, looking for the inefficiencies that institutional capital was missing or ignoring. Most of those experiments lost money. Some of them taught me what actually works. All of them taught me what does not.

I am not a former Goldman Sachs analyst. I do not have a PhD in computer science. I do not come from venture capital or hedge fund operations. What I do have is five years of direct, hands-on engagement with crypto markets at retail scale, specifically in the frontier-market corridors that QAELUM is designed to capture. I have personally watched the inefficiencies that this system targets. I have personally tried to capture them with tools that were not adequate. QAELUM is what I think the adequate tools should look like.

I want to be honest about what this means for investors. A first-time founder with retail-origin experience is a higher-risk profile than a serial founder with institutional pedigree. The risk premium that investors are pricing into this seed round should reflect that reality. I am asking investors to back the architecture and the operational knowledge, not to assume institutional polish I do not yet have.

Why I Am Doing This

I have spent five years watching opportunities slip past me because the right tooling did not exist. I have watched Western quant firms ignore markets that are too small or too regulatorily uncertain for them. I have watched African data providers offer expensive satellite imagery while the actual ground truth was available from people who would happily submit it for fair compensation. The system that connects these dots does not exist yet. I am building it because nobody else seems likely to. If I do not build it, the opportunity I have spent five years observing will pass to someone else who probably will not understand the markets the way I do. That is not acceptable to me.

What I Bring vs What I Need to Add

Bring: Five years of operational engagement, documented architectural specifications, direct contacts in West Africa for QCIS bootstrap, familiarity with failure modes that eliminated previous operators, sustained execution discipline.

Need to add: Technical co-founder (target: late Phase 1 / early Phase 2), senior advisors with Bittensor ecosystem standing, audited operational track record, institutional relationships.

Nahum

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